

IIDA, Japan

WMO: 476370

Lat: 35.5233N

Lon: 137.8225E

Elev: 1735

StdP: 13.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.3	(c) 22.7	(d) 5.3	(e) 7.7	(f) 35.5	(g) 8.6	(h) 9.1	(i) 35.3	(j) 20.6	(k) 41.5	(l) 18.5	(m) 40.9	(n) 2.7	(o) 310	(p) 0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 17.5	(c) 92.1	(d) 74.1	(e) 89.7	(f) 73.3	(g) 87.2	(h) 72.5	(i) 76.2	(j) 86.9	(k) 75.1	(l) 85.0	(m) 74.0	(n) 83.2	(o) 9.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	133.4	79.8	72.5	128.4	78.9	71.5	123.9	78.0	41.0	87.1	39.9	85.0	38.9	83.6	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.5	13.4	15.6	95.7	2.7	1.9	13.7	97.0	12.1	98.2	10.6	99.2	8.6	100.6
			14.2	78.0	2.5	1.3	12.3	78.9	10.9	79.7	9.4	80.5	7.6	81.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	34.4	36.7	43.5	53.8	63.3	70.0	77.2	79.1	71.7	60.5	48.6	39.0	
	DBStd	16.21	4.41	5.48	5.77	6.36	4.63	4.02	4.16	3.60	5.43	5.62	5.73	5.12	
	HDD50	1547	484	373	216	34	0	0	0	0	2	95	343		
	HDD65	4316	949	792	667	339	88	7	0	11	163	493	807		
	CDD50	3950	0	1	14	147	411	599	844	902	652	326	52	2	
	CDD65	1243	0	0	0	2	34	155	379	437	213	23	0	0	
	CDH74	9168	0	0	0	43	344	882	2956	3616	1240	87	0	0	
	CDH80	3235	0	0	0	4	53	170	1164	1489	349	6	0	0	
Wind	WSAvg	4.8	4.7	5.0	5.6	5.7	5.4	4.8	4.9	4.6	4.0	3.7	3.9	4.7	
Precipitation	PrecAvg	60.4	2.4	3.0	4.8	4.9	5.6	6.9	8.0	5.6	7.9	5.6	3.6	2.4	
	PrecMax	81.5	5.6	6.4	9.4	10.1	13.0	14.1	18.1	14.3	13.8	14.6	8.2	5.6	
	PrecMin	39.4	0.2	0.8	2.3	0.8	1.8	2.2	1.5	1.4	2.9	0.6	0.7	0.1	
	PrecStd	11.9	1.3	1.5	1.9	2.4	2.8	2.8	4.7	3.1	3.2	3.0	2.1	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	60.7	67.6	78.7	85.1	87.4	94.7	95.4	90.2	80.1	69.0	59.0	
		MCWB	44.2	47.2	50.5	57.7	62.2	68.8	74.9	75.1	72.4	66.5	57.2	51.2	
	2%	DB	49.4	54.4	62.8	74.0	80.9	84.1	92.0	92.6	87.0	76.4	64.9	54.5	
		MCWB	40.8	43.9	47.5	55.2	61.3	67.5	74.1	74.3	71.2	63.9	54.9	46.6	
	5%	DB	46.2	51.0	58.9	70.0	77.8	81.8	89.5	90.5	84.2	73.1	61.9	51.3	
		MCWB	38.0	41.5	46.5	54.1	60.1	66.5	73.5	73.8	70.1	62.2	53.4	43.7	
	10%	DB	43.3	47.4	55.4	66.2	74.9	79.3	86.7	88.1	81.1	70.2	58.8	48.5	
		MCWB	35.9	39.0	45.5	52.9	59.0	65.8	72.8	73.4	69.0	61.2	51.8	41.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	51.7	55.6	64.1	68.7	73.3	78.1	77.8	74.7	69.6	61.8	53.9
			MCDB	50.2	55.4	58.8	68.4	75.5	81.1	89.3	89.6	83.5	75.1	64.4	56.5
2%		WB	42.0	46.5	52.3	60.3	65.6	71.6	76.3	76.4	73.3	67.3	57.8	48.4	
		MCDB	47.5	52.2	57.4	67.3	72.3	78.0	86.9	87.4	81.3	72.8	61.9	52.2	
5%		WB	39.1	42.7	49.7	57.6	63.7	70.2	75.0	75.4	72.3	65.1	55.1	45.1	
		MCDB	44.7	48.3	56.6	65.3	71.9	76.3	85.0	85.6	79.8	70.7	59.9	49.7	
10%		WB	36.7	40.0	46.7	55.4	61.8	68.7	73.8	74.5	71.1	62.8	52.7	42.6	
		MCDB	42.1	45.9	53.7	63.9	70.6	75.3	83.2	84.0	78.1	68.5	57.4	47.6	
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.8	20.3	21.6	21.0	17.2	16.3	17.5	16.5	17.6	18.7	17.3	
		MCDBR	21.1	24.2	26.4	27.4	26.8	21.4	21.1	20.9	19.8	20.3	22.2	21.2	
		MCWBR	13.3	14.5	14.5	12.1	10.9	8.2	7.0	6.5	7.1	9.5	12.8	13.4	
	5% WB	MCDBR	19.3	20.1	20.7	20.4	19.8	15.3	18.7	18.7	16.4	16.1	17.3	18.5	
		MCWBR	13.5	14.0	14.5	11.2	10.3	7.0	6.7	6.4	6.7	8.7	11.8	13.2	
Clear-Sky Solar Irradiance	taub	0.371	0.419	0.495	0.546	0.558	0.590	0.571	0.568	0.498	0.453	0.397	0.367		
	taud	2.138	2.009	1.864	1.806	1.831	1.824	1.933	1.930	2.081	2.123	2.196	2.205		
	Ebn at Noon	253	253	244	239	239	230	234	231	242	241	243	247		
	Edh at Noon	37	48	60	67	67	67	60	59	48	42	35	33		
All-Sky Solar Radiation	RadAvg	827	1040	1308	1525	1666	1478	1513	1541	1252	1044	830	721		
	RadStd	55	69	89	145	180	124	212	141	112	105	68	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page